

# Industrial Security (Bachelor)

## Exercise Sheet – Section 03 Industrial Network Architectures

### Exercise 3.1 – Draw a Purdue Diagram

A small water-treatment plant has:

- 6 PLCs controlling pumps and valves
- 2 HMI workstations in the control room
- A SCADA server and a Historian
- A patch/AV staging server
- Office IT (email, ERP, web)
- Vendor remote-support access

Draw the full Purdue diagram. Label every level. Show every firewall and the Industrial DMZ explicitly. Mark vendor remote-support clearly – where does it terminate?

### Exercise 3.2 – Zones and Conduits

Using your Exercise 3.1 diagram:

1. Define at least four zones with target Security Levels (SL-T 1–4).
2. For every conduit between zones, list allowed protocols, source, destination, and direction (uni- or bi-directional).
3. Identify one conduit where a data diode would be appropriate. Justify why a stateful firewall is insufficient.

### Exercise 3.3 – Lab: Segmentation

See `labs/lab-03-segmentation`.

1. In GNS3 or EVE-NG, build a virtual network with subnets Enterprise (10.0.0.0/24) and Control (192.168.10.0/24).
2. Place a firewall (pfSense or VyOS) between them, with a DMZ subnet.
3. Allow only Historian replication from Control to DMZ on TCP/5450, and Historian pull from Enterprise to DMZ.
4. Verify that a host in Enterprise cannot reach the Control subnet directly using `nmap`.

#### Caution

Lab only. Never run scans against systems you do not own.

**Exercise 3.4 – Topology Trade-Offs**

Compare star vs. ring topologies on the plant floor in three rows: *failure modes*, *recovery time*, *cabling cost*. Write 2–3 sentences per cell.